

Hatcher Chapter 0 Notes

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1 Retraction, Homotopy, and Deformation Retraction

This part is mainly a review of the definitions I know:

This part is mainly ways of classifying the spaces with similar shapes, with the definition much broader than homeomorphisms (which is too strict).

Definition 1 (Retraction). *Given a subspace $A \subset X$, a **Retraction** of X onto A , is a continuous map $r : X \rightarrow X$, such that $r(X) = A$, and the restriction $r|_A = \text{id}_A$.*

A more formal definition can be viewed as a continuous idempotent map, or $r : X \rightarrow X$ satisfies $r^2 = r$.

Here are some examples:

Example 1. *Every constant map is a trivial example of retraction, since $r : X \rightarrow X$ by $r(x) = x_0$ for all $x \in X$ (given some $x_0 \in X$) is continuous, $r^2 = r$, and is an identity on x_0 itself.*

Example 2. $\mathbb{R}^2 \setminus \{0\}$ has a retraction onto S^1 , by the map $r : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2 \setminus \{0\}$, $r(\bar{x}) = \frac{\bar{x}}{\|\bar{x}\|}$. (In fact, this is coming from a deformation retraction).

However, **Example 1** is an example that retraction is too broad as a definition, since every nonempty space has a retraction to a singleton. So, the classification of “space with similar shape” needs to be broadened.

For this, it requires a notion of “deformation” on continuous maps:

Definition 2 (Homotopy). *A **Homotopy** between two continuous maps $f_0, f_1 : X \rightarrow Y$, is a continuous map $F : X \times I \rightarrow Y$, such that every $x \in X$, one has $f(x, 0) = f_0(x)$, and $f(x, 1) = f_1(x)$.*

Which, for all $t \in [0, 1]$, the map $f_t : X \rightarrow Y$ is defined as $f_t(x) := f(x, t)$.

*Which, we say f, g are **Homotopic**, denoted as $f \simeq g$.*

Which, it's known that homotopy is an equivalence relation on the set of continuous maps.

Now, sometimes homotopy needs extra requirements: For instance, having the homotopy be independent on some subspace. Which leads to the following definition:

Definition 3. Given a subset $A \subseteq X$, $F : X \times I \rightarrow Y$ a homotopy. F is called a **Homotopy Relative to** A , if for any $t \in [0, 1]$, $f_t|_A = f_0|_A$. In other words, the homotopy remains fixed on A .

As an example, one can see the following:

Example 3. The map $r : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2 \setminus \{0\}$ by $r(\bar{x}) = \frac{\bar{x}}{\|\bar{x}\|}$ is in fact homotopic to the identity map; moreover, it's homotopic relative to S^1 . Which, can be constructed as follow:
Let $X := \mathbb{R}^2 \setminus \{0\}$, define $F : X \times I \rightarrow X$ by $F(\bar{x}, t) := \frac{t\bar{x}}{\|\bar{x}\|} + (1-t)\bar{x}$. Which, it's clearly continuous. Also, one has $f_0(\bar{x}) = \bar{x}$ (identity), and $f_1(\bar{x}) = \frac{\bar{x}}{\|\bar{x}\|}$ (r).
Finally, for all $\bar{x} \in S^1$, since $\|\bar{x}\| = 1$, then one has $f_t(\bar{x}) = t\bar{x} + (1-t)\bar{x} = \bar{x}$, it's identity on S^1 .

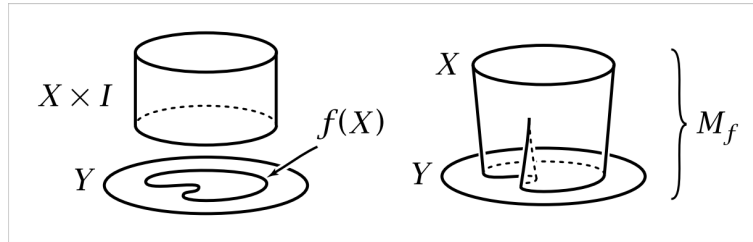
Now, in case to deal with deformation within the space itself, one combines the two definition above:

Definition 4 (Deformation Retraction). Given $A \subseteq X$, a **Deformation Retraction** of a space X onto a subspace A , is a homotopy $F : X \times I \rightarrow X$, such that $f_0 = \text{id}_X$, while $f_1 : X \rightarrow X$ is a retraction onto A . Also, it's required that $f_t|_A = \text{id}_A$ for all $t \in [0, 1]$.

The above **Example 3** shows that r is a deformation retraction, instead of just a retraction. Finally, this gives rise to a new definition:

Definition 5 (Mapping Cylinder). Given a map $f : X \rightarrow Y$, the **Mapping Cylinder** M_f is the quotient space of the disjoint union $(X \times I) \sqcup Y$, obtained by $(x, 1) \sim f(x)$ for all $x \in X$.

Here is an illustration about Mapping Cylinder: Which, the mapping cylinder satisfies some properties:



Proposition 1. The Mapping Cylinder M_f deformation retracts onto the subspace Y .

Proof. Define a continuous map $F : M_f \times I \rightarrow M_f$ as follow:

- For $y \in Y$, define $F(y, s) = y$ for all $s \in [0, 1]$ (which, for all $x \in X$, one has $f(x) \sim (x, 1)$ in M_f).
- For $t \in [0, 1]$ and $s \in [0, 1]$, define $F((x, t), s) = (x, s + (1-s)t)$.

Which, the map is continuous (check Hatcher Appendix A.17, which claims that if $f : X \rightarrow Y$ is a quotient map, with Z being locally compact, then $f \times \text{id}_Z : X \times Z \rightarrow Y \times Z$ is a quotient map). When $s = 0$, one has $F(y, 0) = y$ and $F((x, t), 0) = (x, t)$ for $t < 1$; also for $s = 1$, one has $F(y, 1) = y$ and $F((x, t), 1) = (x, t) = f(x)$. Finally, it's stationary on Y by the definition, hence it deformation retracts onto Y . $\square$

Finally, a relation on the spaces can be given using homotopy:

Definition 6. *Homotopy Equivalence* A continuous map $f : X \rightarrow Y$ is a **Homotopy Equivalence**, if there exists continuous map $g : Y \rightarrow X$, such that $g \circ f \simeq \text{id}_X$, and $f \circ g \simeq \text{id}_Y$. Which, in this case it's said that X and Y are **Homotopy Equivalent**, or having the same **Homotopy Type**, which is denoted as $X \simeq Y$.

Which, here it can give a specific definition of space with stronger connectedness:

Definition 7 (Contractable Space). *A space is called **Contractable**, if it is homotopy equivalent to a singleton.* Which, having the identity map being homotopic to a constant map is sufficient.

Question: Is the above definition an equivalence relation?

As a side note, being contractable may be a bit different from simply-connected space, since there are spaces where it's contractable, but maybe not path-connected.

2 Cell Complexes